700. Search in a BST

Platform LeetCode

Difficulty

Easy Medium Hard

Date 2026-07-29

Column 1

Problem Summary

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Given the root of a BST and a target value, return the subtree rooted at the node with that value. Return null if not found.

Algo

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1. If root == null return null (not found).
2. If root.val == val return root (found).
3. If val < root.val recurse left.
4. Else recurse right.

Time Complexity & Spa

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O(h) — where h is tree height. O(log n) balanced, O(n) skewed.

O(h) — recursion stack depth equals tree height.

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Column 2

Approach

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Recursive BST search — at each node, compare target with root.val and go left or right accordingly. Eliminate half the tree at each step.

Observations

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- O(log n) for balanced BST, O(n) worst case for skewed tree.
- Iterative version possible: while(root != null) — avoids recursion stack.
- This is the building block for insert, delete, and validate BST.

Key Insights

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- BST property guarantees we never need to search both subtrees — one comparison eliminates half the tree.
- Returns the entire **subtree** rooted at the found node, not just the value.
- Base case root == null handles both empty tree and search miss (fell off the tree).

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